

# **Reflection on Applying Cybersecurity Theories to My Career**

A Personal Reflection Paper

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## **Introduction**

This course has given me valuable insights into cybersecurity theories, frameworks, and practices, as well as how they apply in real-world business environments. As an Application Development Analyst in the financial services industry, I must constantly balance security, compliance, and efficiency while developing and maintaining enterprise-level systems. Through this course, I gained practical understanding of the CIA triad, business continuity planning, vulnerability assessments, Zero Trust Architecture, and social engineering countermeasures. These lessons now guide me in approaching projects more strategically and ethically.

## **Applying the CIA Triad**

The Confidentiality, Integrity, and Availability (CIA) Triad is at the heart of cybersecurity, and I apply it daily when constructing and protecting data pipelines for financial portfolios. Platforms such as Azure Synapse protect confidentiality through encryption and access restrictions, while integrity is ensured through hashing, digital signatures, and strong transaction logging. Availability is supported by redundant clusters and failover systems (Stallings, 2018).

One major lesson I took from the readings was that emphasizing one pillar while neglecting others weakens the system. For instance, encryption is ineffective if outages disrupt system availability. This realization has helped me advocate for balanced investment in confidentiality, integrity, and availability, instead of prioritizing only one dimension.

## **Risk Management and Business Continuity**

The module on business continuity planning (BCP) highlighted the importance of resilience in financial IT systems. In my role, downtime can result in millions of dollars in delayed transactions or lost opportunities. According to Swanson et al. (2010), effective continuity planning must be flexible, frequently tested, and formally approved by leadership. I now view BCP as more than a compliance task—it is central to maintaining customer trust and meeting regulatory obligations. This change in perspective has also made me more assertive when discussing backups, disaster recovery strategies, and executive engagement in continuity initiatives.

## **Vulnerability Assessments and Penetration Testing**

Another valuable area of learning was the penetration testing methodology, which includes planning, reconnaissance, scanning, exploitation, and reporting (Caselli & Kargl, 2014). While my primary responsibilities involve writing automation scripts, I often collaborate with penetration testers who evaluate third-party vendor integrations. A colleague at a consulting firm once conducted a red team engagement in which he gained access to a bank's portfolio database and even its vault using fake phishing emails and a forged badge. This exercise demonstrated that vulnerabilities extend across both digital and physical systems.

Because of this, I now insist that vendor code, access controls, and system configurations undergo more rigorous checks before implementation. These lessons emphasize the need to evaluate weaknesses proactively rather than reactively.

## **Zero Trust in Cloud Environments**

This course also reinforced the importance of Zero Trust Architecture (ZTA). Genworth recently transitioned its cloud applications from traditional VPN access to a Zero Trust model. Although initially unpopular due to repeated authentication prompts, ZTA eliminated many long-standing “implicit trust” gaps by enforcing the principle of “never trust, always verify” (Kim et al., 2016). This taught me to view authentication as a balance between usability and security, not as a one-sided constraint.

## **Social Engineering Awareness**

One of the most practical takeaways was the study of social engineering. Despite strong technical safeguards, human error remains the most significant vulnerability (Wang et al., 2020). Phishing simulators at work often reveal gaps in employee awareness. The lectures on dumpster diving, impersonation, and psychological manipulation highlighted the importance of fostering a security-focused culture. As a result, I have contributed to training and awareness initiatives to help colleagues recognize and respond to social engineering attempts more effectively.

## **Ethical Leadership Connection**

Finally, I appreciated the course’s emphasis on servant leadership. In financial services, safeguarding data means protecting clients’ livelihoods, pensions, and investments. Cybersecurity is therefore not only a technical requirement but also a moral responsibility. By adopting frameworks such as the CIA triad and ZTA and supporting training programs, I contribute to

protecting both the organization and the broader community it serves. Leadership in cybersecurity requires making ethical choices while embracing technological innovation.

## Conclusion

This course has enabled me to put theoretical knowledge into practice. The CIA triad, business continuity planning, penetration testing, Zero Trust, and social engineering defenses are no longer abstract ideas but tools I apply daily. Incorporating these frameworks strengthens Genworth's security posture while also shaping me into a more ethical and service-oriented leader. Moving forward, I will continue to use these lessons to design systems that are secure, resilient, and aligned with both organizational and ethical priorities.

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## References

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